

Relational Kernels: Geometry, Causality, and Bell Nonlocality

Kevin Grassi, MD

kmgrassi@gmail.com

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Abstract

We study normalized positive semidefinite (PSD) relational kernels as common-origin structures from which geometric, causal, and quantum correlation properties may be derived. Given a symmetric kernel G , we define an associated geometry via

$$d(i, j) = -\log |G(i, j)|,$$

and show that multiplicative submultiplicativity of $|G|$ is necessary and sufficient for (approximate) metric emergence.

Within the same framework, we define Bell-type correlators $E(a, b) = G(A_a, B_b)$ and analyze the resulting CHSH values. We prove that normalized PSD kernels can reproduce correlations up to the Tsirelson bound while preserving no-signaling.

Our main result establishes a tradeoff between geometric consistency and Bell nonlocality. We prove (Theorem 3.3, machine-checked in Lean 4) that if G satisfies uniform submultiplicativity with parameter $\mu > 0$ over the Bell sector, then $|\text{CHSH}(G)| < 2\sqrt{2}$: maximal Bell violation is incompatible with any nonzero geometric consistency. Numerical optimization over 10^6 random feasible configurations (Appendix C) confirms that the achievable CHSH value decreases monotonically with μ , and characterizing the tight bound $C^*(\mu)$ remains an open problem.

We further construct explicit examples demonstrating *sectoral coexistence*: a single kernel can support approximate geometry on one subset and Bell nonlocality on another.

We conclude that geometry, causality, and nonlocal correlations arise from distinct structural constraints on a common relational substrate, with a precise quantitative tension between geometric transitivity and nonlocality.

1 Introduction

A central question in the foundations of quantum theory is why nonlocal correlations are limited in strength. Bell’s theorem [1] shows that quantum mechanics permits correlations incompatible with any local hidden-variable model, and the CHSH inequality [2] provides a canonical quantitative test: classical theories satisfy $|S| \leq 2$, while quantum mechanics allows violations up to the Tsirelson bound $|S| \leq 2\sqrt{2}$ [3, 4]. However, the no-signaling principle alone permits correlations as large as $|S| = 4$. What structural principle explains the quantum ceiling?

A wide range of answers has been proposed. Information-theoretic principles such as information causality [5] constrain the advantage obtainable from correlations, while frameworks such as the Navascués–Pironio–Acín (NPA) hierarchy [7] characterize quantum correlations through positive semidefinite (PSD) moment matrices. Geometric and functional-analytic approaches, including connections to the Grothendieck inequality [18], further highlight that the set of admissible correlations has a rich underlying structure.

In this work, we pursue a complementary perspective: we study a single class of mathematical objects—normalized positive semidefinite kernels—and show that they simultaneously encode

geometric, causal, and nonlocal properties. Concretely, a PSD kernel G (equivalently, a Gram matrix of unit vectors) can be interpreted in three distinct but related ways:

- as defining a notion of distance via $d(i, j) = -\log |G(i, j)|$,
- as defining Bell-type correlators via $E(a, b) = G(A_a, B_b)$,
- and, in a directed extension, as encoding causal relations.

This unified viewpoint suggests that geometry and nonlocality are not independent features, but arise from different structural constraints imposed on the same relational substrate.

The key structural parameter we introduce is a multiplicative consistency condition on the kernel, which we call μ -submultiplicativity. Informally, this condition enforces transitivity of correlations: if i is strongly related to j , and j to k , then i must retain a controlled level of relation to k . As we show, this condition is precisely what is required for the log-distance $d(i, j) = -\log |G(i, j)|$ to satisfy the triangle inequality, and hence to define a metric (Theorem 3.2). In this sense, geometric structure emerges exactly when correlations compose coherently.

Our main result establishes a tension between this geometric consistency and quantum nonlocality. Specifically, we prove (Theorem 3.3, machine-checked in Lean 4) that if μ -submultiplicativity holds uniformly on the Bell sector with any $\mu > 0$, then the CHSH value is strictly bounded below the Tsirelson limit. Intuitively, maximal Bell violation requires certain correlations to vanish, while geometric consistency forbids such degeneracies by enforcing propagation across relational chains. Thus, maximal nonlocality is incompatible with even a small amount of geometric coherence.

We complement this result with a numerical study (Appendix C), which shows that the maximum achievable CHSH value decreases monotonically as μ increases. This suggests a continuous tradeoff between a “nonlocal” regime, where correlations can be tuned independently, and a “geometric” regime, where they are constrained by transitivity. Determining the exact tradeoff curve remains an open problem.

Importantly, this obstruction is global rather than local. We construct explicit examples in which a single kernel exhibits approximate geometric structure on one subset of indices and near-maximal Bell violation on another (Section 5). Thus, geometry and nonlocality can coexist within the same system, provided the consistency condition is not imposed uniformly.

Finally, we consider directed kernels and show that causal structure is logically independent of the metric structure derived from the symmetric kernel (Section 6). Additional conditions, such as finite-speed propagation, are required to align causal and geometric notions.

Throughout, we distinguish sharply between mathematical results and physical interpretation. While the formal statements concern properties of PSD kernels, the broader motivation is that familiar structures—distance, causality, and quantum correlations—may be understood as different manifestations of a common relational framework.

2 Relational Kernel Framework

Definition 2.1 (Geometric consistency). A normalized PSD kernel G satisfies μ -submultiplicativity if

$$|G(i, k)| \geq \mu \cdot |G(i, j)| \cdot |G(j, k)|$$

for all triples i, j, k in the index set.

Intuitively, μ -submultiplicativity enforces a form of transitivity of correlations: if i is strongly related to j , and j to k , then i must retain at least a fraction μ of that composite relationship to k . The parameter μ therefore quantifies how coherently relationships propagate across chains. Larger μ corresponds to more “geometric” behavior, while small μ permits increasingly fragmented or non-transitive structure.

Definition 2.2 (Intrinsic parameter). The *geometric consistency parameter* of G is

$$\mu(G) := \inf_{i,j,k} \frac{|G(i,k)|}{|G(i,j)||G(j,k)|}.$$

The quantity $\mu(G)$ can be viewed as a worst-case measure of relational consistency across the system. When $\mu(G)$ is close to 1, correlations compose reliably and support a stable notion of distance. When $\mu(G)$ is close to 0, multi-step relationships can degrade arbitrarily, and any geometric interpretation becomes unreliable.

Remark 2.3 (Scope of the condition). The μ -submultiplicativity condition is a strong structural constraint: generic PSD kernels (i.e., Gram matrices of randomly chosen unit vectors) typically do *not* satisfy it with $\mu > 0$. This paper studies the specific subclass of kernels that do. The condition is justified on two grounds. First, it is precisely the right condition for metric emergence: submultiplicativity is *necessary and sufficient* for the log-distance $d(i,j) = -\log |G(i,j)|$ to satisfy the triangle inequality (Theorem 3.2; see Appendix D for the full equivalence). Second, the additive defect $\delta = -\log \mu$ admits four equivalent interpretations—geometric consistency slack, triangle inequality defect, transport fidelity loss, and information retention—all proved equivalent in Lean 4 (Appendix D).

3 Main Results

Theorem 3.1 (Tsirelson bound). *For any normalized PSD kernel G and any choice of Bell indices A_0, A_1, B_0, B_1 ,*

$$|\text{CHSH}(G)| \leq 2\sqrt{2}.$$

This is a standard result [3, 4]; see also [16] for the SDP/Gram-matrix derivation.

Theorem 3.2 (Metric equivalence). *Let G be a normalized PSD kernel with $|G(i,j)| > 0$ for all i,j , and define $d(i,j) = -\log |G(i,j)|$. Then the following are equivalent:*

1. $|G(i,k)| \geq |G(i,j)||G(j,k)|$ for all i,j,k (multiplicative submultiplicativity);
2. $d(i,k) \leq d(i,j) + d(j,k)$ for all i,j,k (triangle inequality).

More generally, μ -submultiplicativity with parameter $\mu \in (0, 1]$ is equivalent to an approximate triangle inequality with additive defect $\delta = -\log \mu$:

$$d(i,k) \leq d(i,j) + d(j,k) + \delta.$$

This result formalizes the idea that geometry is nothing more than consistent composition of relationships. The triangle inequality emerges precisely when indirect connections (via intermediate points) do not artificially shorten distances. Thus, metric structure is not an additional assumption, but a direct consequence of multiplicative coherence in the underlying kernel.

The proof is a direct computation using $-\log$; it is machine-checked in Lean 4 (bundle 6d12681e). See Appendix D for discussion and the four-interpretation equivalence.

Theorem 3.3 (Geometric obstruction). *If $\mu > 0$ holds uniformly on the Bell sector, then $|\text{CHSH}| < 2\sqrt{2}$.*

The content of this theorem is that maximal quantum nonlocality requires a breakdown of relational transitivity. Saturating the Tsirelson bound forces certain correlations to vanish, while μ -submultiplicativity forbids such degeneracies by requiring correlations to propagate across chains. In this sense, geometric consistency and maximal Bell violation represent incompatible structural regimes of the same underlying object.

Proof outline. Any normalized PSD kernel achieving $\text{CHSH} = 2\sqrt{2}$ requires $G(A_0, A_1) = 0$ (by the Tsirelson decomposition: the equality condition $\sqrt{2+2\alpha} + \sqrt{2-2\alpha} = 2\sqrt{2}$ holds iff $\alpha = 0$). But $G(A_0, A_1) = 0$ together with μ -submultiplicativity implies that for each b , either $G(A_0, B_b) = 0$ or $G(A_1, B_b) = 0$. This forces at least two correlators to vanish, giving $|\text{CHSH}| \leq 2$. Contradiction. The full argument is machine-checked in Lean 4; see Appendix A.6. $\square$

Remark 3.4 (Quantitative tradeoff). Numerical optimization (Appendix C) over 10^6 random feasible configurations with full 4×4 PSD eigenvalue verification shows that the maximum achievable $|\text{CHSH}|$ decreases monotonically with μ : from ≈ 2.82 at $\mu = 0$ (consistent with Tsirelson) to ≈ 2.34 at $\mu = 1$. Determining the tight bound $C^*(\mu) := \sup\{|\text{CHSH}(G)| : G \text{ is } \mu\text{-submultiplicative on } T\}$ analytically is an open problem. A natural candidate is $4/\sqrt{2+\mu}$, derived from a symmetric-extremizer ansatz (Appendix A), but the numerical data shows this formula *underestimates* $C^*(\mu)$ for $\mu > 0$: the actual extremizers are asymmetric.

4 Analysis of the CHSH–Geometry Tradeoff

We analyze the structure of the optimization problem underlying the geometry–nonlocality tradeoff.

Step 1: PSD constraints. For any unit-vector representation $G(i, j) = \langle v_i, v_j \rangle$ [3, 16], define

$$x_{ab} := G(A_a, B_b).$$

Using Gram structure and Cauchy–Schwarz (cf. [4, 16]),

$$(x_{00} + x_{01})^2 \leq 2(1 + G(B_0, B_1)),$$

and analogous inequalities for all permutations.

Step 2: Handling all sign configurations. Define

$$S := x_{00} + x_{01} + x_{10} - x_{11}.$$

We bound $|S|$ by considering extremal configurations. Without loss of generality, we reduce to maximizing S .

Key observation: The CHSH optimum occurs when $G(B_0, B_1) \leq 0$.

Proof sketch of reduction: If $G(B_0, B_1) > 0$, one can flip $v_{B_1} \mapsto -v_{B_1}$, which preserves PSD and submultiplicativity while increasing S . Thus the extremum lies in the sector $G(B_0, B_1) \leq 0$.

Step 3: Apply submultiplicativity. Using

$$|G(B_0, B_1)| \geq \mu |x_{00}| |x_{01}|,$$

and similarly across all triples, we obtain lower bounds on negative correlations.

Step 4: Optimization problem. We maximize $S = x_{00} + x_{01} + x_{10} - x_{11}$ subject to PSD constraints (Gram matrix positive), submultiplicativity inequalities, and $|x_{ab}| \leq 1$.

Conceptually, this optimization captures a competition between two tendencies: maximizing Bell violation favors sharp, highly selective correlations, while μ -submultiplicativity enforces smooth propagation across relational paths. The feasible region therefore interpolates between a “quantum-extremal” regime ($\mu \approx 0$) and a “geometric” regime (μ close to 1). The resulting tradeoff curve quantifies how these competing constraints balance.

Symmetric-extremizer ansatz. A natural conjecture is that the maximum occurs at

$$x_{00} = x_{01} = x_{10} = x, \quad x_{11} = -x,$$

motivated by sign-relabeling symmetry ($A_a \mapsto -A_a$, $B_b \mapsto -B_b$) and permutation equivalence of the CHSH form. Under this ansatz, $|S| \leq 4/\sqrt{2+\mu}$ (Appendix A). However, numerical optimization (Appendix C) reveals that the actual extremizers are *asymmetric*, achieving $|S|$ values above $4/\sqrt{2+\mu}$ while remaining below the Tsirelson bound. The tight bound $C^*(\mu)$ remains open.

The observed monotonic decrease reflects a simple structural tension: enforcing consistency across triples limits the extent to which correlations can be independently tuned. As μ increases, the system becomes increasingly constrained by transitivity, reducing the freedom required to achieve large Bell violations. This provides a quantitative bridge between geometric coherence and nonlocal strength.

Step 5: Ansatz bound. Substituting the symmetric ansatz into constraints gives

$$(2 + \mu)x^2 \leq 1,$$

yielding $|S| = 4x \leq 4/\sqrt{2+\mu}$. As noted, this is a lower bound on $C^*(\mu)$, not the tight bound.

5 Sectoral Coexistence

We provide an explicit kernel demonstrating that geometric consistency and Bell nonlocality can coexist in different sectors of a single kernel.

Let $\{v_i\}_{i \in \mathcal{G}}$ form a near-metric configuration (e.g., Euclidean embedding), let $\{w_j\}_{j \in \mathcal{B}}$ form a Tsirelson-optimal Bell configuration, and define cross-terms to be small: $\langle v_i, w_j \rangle = \epsilon$.

Then the combined Gram matrix remains PSD for sufficiently small ϵ , with the following properties:

- On $\mathcal{G}$: submultiplicativity holds with $\mu > 0$.
- On $\mathcal{B}$: CHSH = $2\sqrt{2}$ up to $O(\epsilon)$ corrections.

Thus a single kernel exhibits both behaviors.

This construction shows that the obstruction identified in Theorem 3.3 is inherently global rather than local. Geometry and strong nonlocality can coexist within a single kernel, provided the consistency condition is not enforced uniformly across all triples. Thus, the tension arises from imposing a single structural regime across the entire system, not from an intrinsic incompatibility at the level of individual correlations.

6 Causality

We formalize directed kernels $K(i \rightarrow j)$ satisfying positivity, the composition inequality $K(i \rightarrow k) \geq \sum_j K(i \rightarrow j) K(j \rightarrow k)$, and acyclicity. Define causal order by

$$i \prec j \iff K(i \rightarrow j) > 0.$$

Theorem 6.1 (Causal partial order). *If K satisfies the semigroup composition inequality and acyclicity, then $\prec$ is a strict partial order (irreflexive, asymmetric, transitive).*

Theorem 6.2 (Metric-causal independence). *Metric structure derived from G does not determine the causal order induced by K .*

This result highlights that geometric proximity alone does not determine directional influence. Even when distances between elements are fixed, multiple incompatible causal orderings may be realized. In particular, causal structure requires additional asymmetric information that is not captured by the symmetric kernel underlying the metric.

Proof. By explicit counterexample on Fin 3: we construct two directed kernels K_1, K_2 that induce incompatible causal orders $\prec_1 \neq \prec_2$ while the symmetric kernel $G(i, j) = \frac{1}{2}(K(i \rightarrow j) + K(j \rightarrow i))$ yields the same metric d in both cases. The construction further shows that any isometry-invariant procedure for assigning causal order from the metric alone must fail, since the two-point metric space admits two distinct strict total orders related by the swap isometry. Machine-checked in Lean 4 (bundles `d4d3c17b`, `5c0ee01b`). $\square$

Theorem 6.3 (Finite-speed sufficiency). *If in addition to the semigroup and acyclicity conditions, the directed kernel satisfies a finite-speed propagation constraint— $K(i \rightarrow j) > 0$ implies $d(i, j) \leq c \cdot \tau(i, j)$ for some speed $c > 0$ and time-separation function τ —then metric and causal structures are compatible: causal diamonds $\{k : i \prec k \prec j\}$ are bounded in d .*

The finite-speed condition plays the role of a compatibility bridge between geometry and causality. By limiting how quickly influence can propagate relative to distance, it aligns the directed structure with the metric one. Without such a constraint, causal relations can violate geometric intuition, as demonstrated by explicit counterexamples.

Remark 6.4. Without finite-speed propagation, metric–causal compatibility can fail: an explicit counterexample on Fin 3 shows $d(0, 1) = 10 > d(0, 2) = 1$ while $1 \in \Diamond(0, 2)$ (bundle `d4d3c17b`, Theorem 6). Thus finite-speed propagation is not merely a physical assumption but a mathematical necessity for compatibility.

7 Discussion

The results of this paper admit a clean structural interpretation: geometry corresponds to transitivity of correlations, while nonlocality corresponds to the violation of transitivity.

Interpretation (heuristic). The parameter μ may be interpreted as a measure of compositional stability of correlations, analogous to contraction coefficients in data-processing inequalities [20] and to the role of metric constraints in the correlation polytope literature [8, 9]. The connection between $-\log |G|$ and approximate metric structure is grounded in Schoenberg’s classical theory linking positive definite kernels to metric embeddings [15, 19]. No physical mechanism is assumed.

From this perspective, geometry corresponds to the stability of correlations under composition, while nonlocality reflects the possibility of bypassing such compositional constraints. The parameter μ provides a quantitative measure of this stability, interpolating between a fully transitive (geometric) regime and a maximally nonlocal one. The tradeoff identified in this work can thus be understood as a tension between compositional coherence and correlation extremality.

8 Conclusion

We have established:

1. A rigorous no-go theorem (Theorem 3.3, machine-checked): geometric consistency ($\mu > 0$) is incompatible with maximal Bell violation.
2. Numerical evidence for a monotone tradeoff $C^*(\mu)$ between geometry and nonlocality, with the tight bound remaining an open problem.
3. The necessity-and-sufficiency of submultiplicativity for metric emergence (Theorem 3.2, machine-checked), with four equivalent characterizations of the defect parameter.
4. Explicit coexistence constructions showing the obstruction is global, not sector-local.
5. Formal independence of metric and causal structures (Theorems 6.2–6.3, machine-checked), with finite-speed propagation identified as the bridge condition.

Open problems include: determining the tight bound $C^*(\mu)$ analytically (see Appendix A.5), extension to multipartite scenarios, and physical realization of μ .

We emphasize that interpretations relating this framework to spacetime or physical systems remain speculative. See [17] for a comprehensive review of Bell nonlocality, and [5, 6, 18] for alternative principles bounding nonlocal correlations.

At a conceptual level, these results suggest that familiar structures such as distance, causal order, and quantum correlations may be viewed as different facets of a common relational substrate. Geometry emerges when correlations compose consistently, while nonlocality appears when this compositional structure is partially relaxed. Understanding how these regimes coexist and transition between one another remains an important direction for future work.

A Symmetric Ansatz Derivation

This appendix records the derivation of the bound $|\text{CHSH}(G)| \leq 4/\sqrt{2+\mu}$ under the symmetric-extremizer ansatz. As shown in Appendix C, this ansatz is not tight: the actual extremizers are asymmetric for $\mu > 0$.

A.1 Setup

Let $x_{ab} := G(A_a, B_b)$ for $a, b \in \{0, 1\}$, and let $S := x_{00} + x_{01} + x_{10} - x_{11}$. Since G is PSD and normalized, there exist unit vectors with $G(i, j) = \langle v_i, v_j \rangle$, so all Gram-matrix principal minors are nonnegative and $|x_{ab}| \leq 1$.

Define $y := G(B_0, B_1)$ and $z := G(A_0, A_1)$. From PSD for pairs and triples, one obtains (equivalently by Cauchy–Schwarz):

$$(x_{00} + x_{01})^2 \leq 2(1 + y), \quad (x_{10} - x_{11})^2 \leq 2(1 - y), \quad (1)$$

$$(x_{00} + x_{10})^2 \leq 2(1 + z), \quad (x_{01} - x_{11})^2 \leq 2(1 - z). \quad (2)$$

A.2 Incorporating geometric consistency

On the Bell sector $T = \{A_0, A_1, B_0, B_1\}$, assume

$$|G(i, k)| \geq \mu |G(i, j)| |G(j, k)|, \quad \mu \in [0, 1].$$

In particular,

$$|y| \geq \mu |x_{00}| |x_{01}|, \quad |y| \geq \mu |x_{10}| |x_{11}|, \quad (3)$$

$$|z| \geq \mu |x_{00}| |x_{10}|, \quad |z| \geq \mu |x_{01}| |x_{11}|. \quad (4)$$

These inequalities enforce a nontrivial lower bound on cross-setting consistency whenever the x_{ab} are large.

A.3 Reduction to symmetric extremal form

The feasible set (PSD constraints + submultiplicativity + box constraints) is compact. Hence a maximizer of $|S|$ exists.

The symmetric-extremizer ansatz posits that the maximum occurs at

$$x_{00} = x_{01} = x_{10} = x, \quad x_{11} = -x,$$

with $x \geq 0$. Under this ansatz, $|S| = 4x$.

Motivation. The CHSH functional $S = x_{00} + x_{01} + x_{10} - x_{11}$ has a $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry under sign relabelings ($A_a \mapsto -A_a$, $B_b \mapsto -B_b$), which preserves both PSD structure and

submultiplicativity. The three positive-sign correlators play symmetric roles in S , and the single negative-sign correlator is distinguished only by its sign. This motivates equating the three positive terms and anti-equating the negative term.

Status. Numerical optimization (Appendix C) shows this ansatz does *not* capture the true extremizers for $\mu > 0$: asymmetric configurations achieve higher $|S|$.

For this symmetric sector, the PSD constraints reduce to

$$2x^2 \leq 1 + |y|,$$

while submultiplicativity gives $|y| \geq \mu x^2$. Combining,

$$2x^2 \leq 1 + \mu x^2 \implies (2 + \mu)x^2 \leq 1.$$

Therefore

$$|S| = 4x \leq \frac{4}{\sqrt{2 + \mu}}.$$

This yields $4/\sqrt{2 + \mu}$ as a bound under the symmetric ansatz.

A.4 Consistency checks

- $\mu = 0$: $|\text{CHSH}| \leq 4/\sqrt{2} = 2\sqrt{2}$ (Tsirelson).
- $\mu > 0$: strict reduction below Tsirelson.
- Monotonicity: $C(\mu) = 4/\sqrt{2 + \mu}$ is strictly decreasing.

A.5 Status of the symmetric ansatz

Numerical optimization (Appendix C) reveals that the symmetric ansatz does *not* capture the true extremizers: for $\mu > 0$, asymmetric configurations achieve $|S|$ values strictly above $4/\sqrt{2 + \mu}$ while remaining below the Tsirelson bound. Thus $4/\sqrt{2 + \mu}$ is a *lower* bound on the true ceiling $C^*(\mu)$, not the tight bound.

Determining $C^*(\mu)$ analytically remains open. Natural approaches include:

1. *SDP relaxation.* Formulate the constrained optimization as a semidefinite program and extract the tight bound from the dual.
2. *KKT analysis on the full (non-symmetric) feasible set.*

We note that the qualitative no-go (Theorem 3.3) does not depend on any specific bound formula and is fully rigorous.

A.6 Supplementary theorem artifacts and downloads

To support reproducibility and peer review, theorem-level supporting artifacts (Lean 4 formalizations, derivation notes, and validation runs) were generated using the Aristotle proof assistant (<https://harmonic.fun>). A curated index of these artifacts is maintained in the public repository at:

https://github.com/kmgrassi/shape-of-correlations/blob/main/data/PROOF_INDEX.md

Current theorem-to-artifact mapping:

- Theorem 3.1 (Tsirelson bound): bundle 3eb07f9d (Tsirelson.lean).
- Theorem 3.3 (Geometric no-go): bundles 3eb07f9d (Obstruction.lean), 269455a4 (CHSHBounds.lean), 22b60d0a (Theorems.lean, Theorem 11—structural form).
- Quantitative bound analysis (Remark 3.4): bundle 42c716ff (CHSH.lean).
- Theorem 3.2 (Metric equivalence): bundle 6d12681e (CorrGeom/MetricEmergence.lean, Theorem B2).

- Four-interpretation equivalence (Appendix D): bundle 8e1b39a8 (Equivalences.lean).
- Theorems 6.1–6.3 (Causality): bundles d4d3c17b (CoreTheorems.lean, Compatibility.lean) and 5c0ee01b (Negative.lean, Positive.lean).
- Metric-emergence support: bundles eba46516 (EmergentMetric.lean) and route bridge bundles 1b6667c0/e47f7af7.
- Bell-local/no-signaling: bundle 399e8153 (EmergentLocalityBell.lean).
- Common-origin coexistence: bundles 0c8103cf (Coexistence.lean) and 5db6ab1b (RelationalKernel.lean).

Public repository. All theorem artifacts are available at <https://github.com/kmgrassi/shape-of-correlations>, with Lean bundles under `data/theorems/` organized by Aristotle request ID.

B Coexistence Construction Details

We give a constructive family showing sectoral coexistence of approximate geometry and strong Bell nonlocality.

B.1 Block-Gram construction

Partition indices into geometric and Bell sectors: $S = S_G \sqcup S_B$. Choose unit-vector families

$$\{u_i\}_{i \in S_G} \subset \mathbb{R}^{d_G}, \quad \{w_j\}_{j \in S_B} \subset \mathbb{R}^{d_B},$$

with Gram blocks $G_G(i, i') = \langle u_i, u_{i'} \rangle$ and $G_B(j, j') = \langle w_j, w_{j'} \rangle$.

Assume G_G satisfies submultiplicativity on S_G with parameter $\mu_G > 0$, and selected $A_0, A_1, B_0, B_1 \in S_B$ realize a Bell-optimal pattern (Tsirelson-level in the $\mu = 0$ limit).

Embed both sectors orthogonally and add weak coupling ϵ :

$$\tilde{u}_i := (u_i, 0), \quad \tilde{w}_j := (\epsilon q_j, \sqrt{1 - \epsilon^2} w_j),$$

where $q_j \in \mathbb{R}^{d_G}$ are unit vectors. Then $G(i, j) = \langle \tilde{u}_i, \tilde{w}_j \rangle$ defines a normalized PSD kernel on all of S .

B.2 PSD and normalization

By construction, G is a Gram matrix of unit vectors, hence automatically PSD and normalized: $G(s, s) = 1$ for all $s \in S$. No additional PSD check is needed beyond explicit embedding.

B.3 Behavior on each sector

Geometric sector S_G . Restricting to S_G , the kernel is exactly G_G , so geometric consistency $\mu_G > 0$ is preserved.

Bell sector S_B . For $j, j' \in S_B$:

$$G(j, j') = (1 - \epsilon^2) \langle w_j, w_{j'} \rangle + \epsilon^2 \langle q_j, q_{j'} \rangle.$$

Hence Bell correlators are

$$E_{ab}(\epsilon) = (1 - \epsilon^2) E_{ab}^{(0)} + \epsilon^2 R_{ab},$$

with $|R_{ab}| \leq 1$. Therefore

$$\text{CHSH}(\epsilon) = (1 - \epsilon^2) \text{CHSH}^{(0)} + \epsilon^2 R, \quad |R| \leq 4,$$

so $\text{CHSH}(\epsilon) = \text{CHSH}^{(0)} + O(\epsilon^2)$. If $\text{CHSH}^{(0)} = 2\sqrt{2}$, then for sufficiently small ϵ , Bell nonlocality remains near-maximal.

B.4 Explicit finite-dimensional toy instance

Take $S_B = \{A_0, A_1, B_0, B_1\}$ and choose vectors in $\mathbb{R}^2$:

$$A_0 = (1, 0), \quad A_1 = (0, 1), \quad B_0 = \frac{1}{\sqrt{2}}(1, 1), \quad B_1 = \frac{1}{\sqrt{2}}(1, -1).$$

Then

$$E_{00} = E_{01} = E_{10} = \frac{1}{\sqrt{2}}, \quad E_{11} = -\frac{1}{\sqrt{2}},$$

so $\text{CHSH} = 2\sqrt{2}$. Combining this Bell block with any geometric block G_G using the ϵ -coupled embedding above, PSD is preserved automatically and coexistence follows.

B.5 Interpretation

This construction proves existence of kernels where one subset exhibits transitivity-compatible geometry ($\mu > 0$) and another subset exhibits strong Bell nonlocality. Thus the obstruction is global-uniform, not sector-local: a single relational substrate can host both behaviors provided constraints are not imposed uniformly across all triples.

C Numerical Optimization Study

We performed a Monte Carlo optimization study to characterize the achievable CHSH values under μ -submultiplicativity.

C.1 Protocol

For each value of $\mu \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$, we sampled configurations $(x_{00}, x_{01}, x_{10}, x_{11}, y, z) \in [-1, 1]^6$ using three phases: (1) uniform rejection sampling (5×10^5 trials), (2) targeted sampling near the symmetric ansatz (3×10^5 trials), and (3) fine-grid exploration near predicted optima (2×10^5 trials), for a total of 10^6 trials per μ value. Feasibility was verified by:

1. *Full PSD check*: all eigenvalues of the 4×4 Gram matrix ≥ 0 ;
2. *Full submultiplicativity*: $|G(i, k)| \geq \mu |G(i, j)| |G(j, k)|$ for all $4^3 = 64$ ordered triples from $\{A_0, A_1, B_0, B_1\}$;
3. *Box constraints*: $|x_{ab}| \leq 1, |y| \leq 1, |z| \leq 1$.

C.2 Results

μ	$4/\sqrt{2+\mu}$	Max $ S $ observed	Symmetric ansatz	Feasible points
0.00	2.828	2.822	consistent	308,803
0.01	2.821	2.822	consistent	285,909
0.05	2.794	2.820	consistent	212,744
0.10	2.760	2.799	mixed	164,994
0.20	2.697	2.762	asymmetric	129,944
0.50	2.530	2.599	asymmetric	78,513
1.00	2.309	2.339	asymmetric	36,158

C.3 Key findings

The qualitative tradeoff is confirmed. In all cases, the maximum observed $|S|$ is strictly below $2\sqrt{2} \approx 2.828$ for $\mu > 0$, and decreases monotonically with μ . This is consistent with Theorem 3.3.

The symmetric-extremizer ansatz is incorrect for $\mu > 0$. For $\mu \geq 0.1$, the observed maxima exceed $4/\sqrt{2+\mu}$, and the maximizing configurations are *asymmetric*: the three positive correlators (x_{00}, x_{01}, x_{10}) take distinct values. This means $4/\sqrt{2+\mu}$ is a lower bound on $C^*(\mu)$, not the tight bound.

The tight bound $C^*(\mu)$ is an open problem. The numerical data suggests $C^*(\mu)$ decreases more slowly than $4/\sqrt{2+\mu}$. Determining the analytic form of $C^*(\mu)$ likely requires solving the full nonlinear program without symmetry reduction, e.g., via SDP relaxation.

D Geometry: Necessity, Sufficiency, and Interpretations

This appendix collects the machine-checked results on the relationship between submultiplicativity and metric structure, drawn from Lean 4 formalizations in bundles `6d12681e`, `eba46516`, and `8e1b39a8`.

D.1 Necessity and sufficiency (Theorem 3.2)

The equivalence of submultiplicativity and the triangle inequality is immediate from the $-\log$ transform:

$$|G(i, k)| \geq |G(i, j)| |G(j, k)| \iff -\log |G(i, k)| \leq -\log |G(i, j)| - \log |G(j, k)|.$$

The key point is that this is an *if and only if*: no weaker multiplicative condition on G suffices for the metric triangle inequality, and conversely the triangle inequality implies the multiplicative bound. This is machine-checked as Theorem B2 in bundle `6d12681e`, which also provides an explicit counterexample on Fin 3 showing that generic PSD kernels (without submultiplicativity) violate the triangle inequality.

D.2 Four equivalent interpretations of μ

The parameter $\delta = -\log \mu$ admits four equivalent characterizations, all proved equivalent in bundle `8e1b39a8`:

1. *Geometric consistency slack*: $|G(i, k)| \geq \mu |G(i, j)| |G(j, k)|$.
2. *Triangle inequality defect*: $d(i, k) \leq d(i, j) + d(j, k) + \delta$.
3. *Transport fidelity*: correlation loss along any two-step path is bounded by factor μ .
4. *Information retention*: mutual information along relational chains degrades by at most δ per step.

The equivalence (1) $\Leftrightarrow$ (2) is the content of Theorem 3.2. Interpretations (3) and (4) follow by viewing $|G(i, j)|$ as a fidelity measure and $-\log |G(i, j)|$ as an information-loss measure, respectively. The formal equivalence is stated and proved in `8e1b39a8/Equivalences.lean`.

D.3 Counterexamples to automatic submultiplicativity

Submultiplicativity is not automatic for PSD kernels. Bundle `6d12681e` (Theorem B1) and bundle `1b6667c0` (Route B, Counterexample B2) exhibit explicit normalized PSD kernels on Fin 3 with $G(0, 1) = G(1, 2) = 0.9$ and $G(0, 2) = 0.1$, violating $|G(0, 2)| \geq |G(0, 1)| |G(1, 2)| = 0.81$. Thus the condition isolates a geometrically meaningful subclass, not a generic property.

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